

ETHAN LIU

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Education

University of Maryland, College Park

Bachelor of Science in Computer Science

GPA: 3.73/4.0

College Park, MD

Expected May 2028

Relevant Coursework

Intro to Computer Systems, Object-Oriented Programming II, Algorithms, Programming Language Technologies and Paradigms, Organization of Programming Languages, Introduction to Data Science, Introduction to Machine Learning, Programming Handheld Systems

Technical Skills

Languages: C, Python, Java, TypeScript, JavaScript, Swift, SQL, HTML/CSS, Rust, Haskell, OCaml, x86 Assembly

Frameworks/Libraries: Pandas, SciPy, scikit-learn, React, Next.js, SwiftUI, Tailwind CSS, JUnit

Developer Tools: Git/GitHub, Eclipse, VS Code, Jupyter Notebook, Xcode, Replit

Projects

UMD Course Recommender

Next.js, React, TypeScript, Tailwind CSS

- Collaborated in a 4-person team using Git to develop a course recommendation application, integrating REST APIs for live UMD course, section, seat availability, and professor data.
- Designed and implemented a client-side TypeScript parser that transforms raw UMD uAchieve degree-audit HTML into structured JSON representing completed courses, in-progress sections, and outstanding requirements.
- Extracted and categorized unfulfilled degree requirements from audit data, including general education status and requirement groups, to support personalized course recommendations.

Diabetes Risk Prediction from CDC Health Indicators

Python, Pandas, SciPy, scikit-learn

- Cleaned and validated a 70,692-record CDC health survey dataset using Python and Pandas, resolving corrupted features and missing values for a 5-member machine learning project.
- Conducted statistical analysis across 21 health indicators using chi-squared, two-sample t-test, and Mann-Whitney U tests to identify BMI, blood pressure, cholesterol, and age as significant predictors of diabetes.
- Authored the primary analysis write-up for logistic regression and support vector machine models, translating results into clear statistical conclusions for a collaborative technical report.

GroupThink

Swift, SwiftUI, Firebase Firestore

- Engineered a native iOS productivity application in SwiftUI using Apple's Screen Time APIs to support on-device application blocking and customizable whitelists during group study sessions.
- Used Git to collaboratively test, debug, and refine new features through an iterative development process, including real-time Firestore synchronization for session lobbies, participant presence, and group settings.

Relevant Experience

Casa Chirilagua & InterVarsity USA

June 2026 – July 2026

Web Strategy & Design Volunteer

Alexandria, VA

- Analyzed a professional website audit and researched nonprofit UX patterns to identify improvements in messaging clarity, donor engagement, and visual accessibility for an organization serving 100+ Latino families.
- Created homepage UI concepts and prototypes informed by peer nonprofit research, with recommendations adopted by the development team as the basis for a full visual redesign.
- Collaborated with the executive director and development manager to incorporate feedback and align design decisions with organizational goals and community-facing services.

Organizations

CS Base @ UMD

November 2025 – Present

Co-Founder & Operations Director

College Park, MD

- Co-founded the University of Maryland chapter of CS Base, a nonprofit organization focused on expanding access to computer science education in traditionally underserved communities.
- Collaborated with Prince George's County schools to develop a strategic plan for introductory Python, Java, and AI/Machine Learning classes for high school students.